

ECON2050 Tutorial 2

Basic Real Analysis II

Semester 2 2026

Xavier Xi Zhang

School of Economics, The University of Queensland

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Announcement

- ▶ Assignment 1 will be released shortly.
- ▶ Due date: 4/09/2026 4:00 pm.
- ▶ Worth 16% of your final grade!
- ▶ I'll go through some important points for the assignment at week 5 tutorial.
- ▶ Feel free to come to my consultation.
 - ▶ My consultation hours are every Thursday from 12:30 pm to 2:00 pm, in 39-125A.

Today's Plan

- ▶ Recap: key ideas from Lecture 2
- ▶ Tutorial questions: Q1–Q5. Hopefully, we can cover all of them...

Recap: Functions of Several Variables

General form

A function of several variables can be written as $f: D \rightarrow \mathbb{R}^m$, where the domain is some set $D \subseteq \mathbb{R}^k$.

Note. A function assigns each input exactly one output!!!

Graph

For $f: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$, the graph is

$$\text{graph}(f) = \{(\mathbf{x}, y) \in D \times \mathbb{R} \subseteq \mathbb{R}^3 : y = f(\mathbf{x})\}.$$

Level curve

The level curve of level c is

$$\{\mathbf{x} \in D : f(\mathbf{x}) = c\}.$$

A contour plot shows level curves for several values of c .

Recap: Continuity

Sequential definition

Let $D \subseteq \mathbb{R}^k$. A function $f: D \rightarrow \mathbb{R}^m$ is **continuous at** $\mathbf{x} \in D$ if for every sequence $(\mathbf{x}_n)$ in D such that $\mathbf{x}_n \rightarrow \mathbf{x}$, we have $f(\mathbf{x}_n) \rightarrow f(\mathbf{x})$.

Note.

- ▶ You may have seen other definitions of continuity. They are all equivalent.
- ▶ Intuition: continuity means a function behaves smoothly, without any sudden jumps.

Continuity on a domain

f is **continuous** if it is continuous at every $\mathbf{x} \in D$.

Recap: Tools for Proving Continuity

Algebra of continuous functions

If $f: D \rightarrow \mathbb{R}$ and $g: E \rightarrow \mathbb{R}$ are continuous at $\mathbf{x} \in D \cap E$, then

$$f + g, \quad f - g, \quad f \cdot g,$$

and f/g when $g(\mathbf{x}) \neq 0$, are continuous at $\mathbf{x}$.

Composition

If $f: D \rightarrow \mathbb{R}^m$ is continuous at $\mathbf{x}$ and $g: E \rightarrow \mathbb{R}^\ell$ is continuous at $f(\mathbf{x})$, with $f(D) \subseteq E$, then

$$g \circ f: D \rightarrow \mathbb{R}^\ell, \quad \mathbf{x} \mapsto g(f(\mathbf{x}))$$

is continuous at $\mathbf{x}$.

Recap: Vector-Valued Functions and Projections

Componentwise continuity

Let $f = (f_1, \dots, f_m) : D \rightarrow \mathbb{R}^m$. Then f is continuous at $\mathbf{x} \in D$ if and only if every component function f_i is continuous at $\mathbf{x}$.

Projection functions

For each $i = 1, \dots, k$, the projection

$$\text{pr}_i : \mathbb{R}^k \rightarrow \mathbb{R}, \quad \mathbf{x} \mapsto x_i$$

is continuous.

Recap: Linear and Affine Functions

Inner product

$$\text{For } \mathbf{x}, \mathbf{y} \in \mathbb{R}^k, \mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^k x_i y_i = x_1 y_1 + \cdots + x_k y_k.$$

Linear and affine functions

- ▶ A function $f: \mathbb{R}^k \rightarrow \mathbb{R}$ is **linear** if there exists $\mathbf{a} \in \mathbb{R}^k$ such that $f(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x}$.
- ▶ It is **affine** if there exist $\mathbf{a} \in \mathbb{R}^k$ and $b \in \mathbb{R}$ such that $f(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x} + b$.

Change formula

For an affine function $f(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x} + b$, if $\Delta \mathbf{x} = \mathbf{x} - \mathbf{x}_0$, and $\Delta y = f(\mathbf{x}) - f(\mathbf{x}_0)$, then

$$\Delta y = \mathbf{a} \cdot \Delta \mathbf{x} = \sum_{i=1}^k a_i \Delta x_i.$$

Question 1

Let f denote an arbitrary function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$. For each statement, determine whether it is true or false. If false, provide an explanation or a counterexample.

- (a) If $f(x_0, y_0) = f(x_1, y_1)$, then $x_0 = x_1$ and $y_0 = y_1$.
- (b) If $a \in \mathbb{R}$, then $f(ax, ay) = a^2 f(x, y)$.
- (c) A vertical line, parallel to the z -axis, can cross the graph of f at most once.
- (d) Two level curves of f can intersect.

Tutorial Q1: True or False

Knowledge used

- ▶ A function assigns each input exactly one output.
- ▶ The graph of $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ contains exactly one height $z = f(x, y)$ above each input (x, y) .
- ▶ A level curve of level C is $\{(x, y) : f(x, y) = C\}$.

Solution

- (a) **False.** Let $f(x, y) = xy$, $(x_0, y_0) = (1, 2)$ and $(x_1, y_1) = (-1, -2)$. Then $f(1, 2) = 2 = f(-1, -2)$, but the two inputs are different.
- (b) **False.** Let $f(x, y) = x + y$. Then
$$f(ax, ay) = ax + ay = a(x + y) = af(x, y),$$
which is not generally equal to $a^2 f(x, y)$.
- (c) **True.** For each (x, y) , the function gives a unique value $z = f(x, y)$.
- (d) **False.** If two distinct level curves $C_1 \neq C_2$ intersected at (x, y) , then $f(x, y) = C_1$ and $f(x, y) = C_2$, a contradiction.

Question 2

Recall that a level curve of a function of two variables f is a subset of f 's domain whose elements are the points (x, y) for which, for some $k \in \mathbb{R}$, $f(x, y) = k$.

Determine the domain on which the following functions can be defined and draw a few level curves for each.

(a) $f(x, y) = (x - 1)^2 + (y + 2)^2$.

(b) $f(x, y) = x + y^2 + 1$.

Tutorial Q2(a): Domain and Level Curves

Knowledge used

- ▶ The domain is the largest set on which the formula is well-defined.
- ▶ A level curve of level k solves $f(x, y) = k$.

Solution

For

$$f(x, y) = (x - 1)^2 + (y + 2)^2,$$

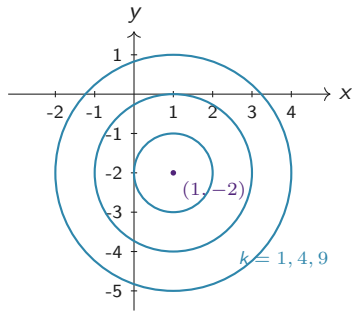
the domain is

$$D = \mathbb{R}^2.$$

The level curve of level k is

$$(x - 1)^2 + (y + 2)^2 = k.$$

For $k \geq 0$, this is a circle centered at $(1, -2)$ with radius $\sqrt{k}$.



Tutorial Q2(b): Domain and Level Curves

Knowledge used

- ▶ Level curves are found by setting $f(x, y) = k$ and solving for a geometric relation.
- ▶ Different values of k shift the level curve.

Solution

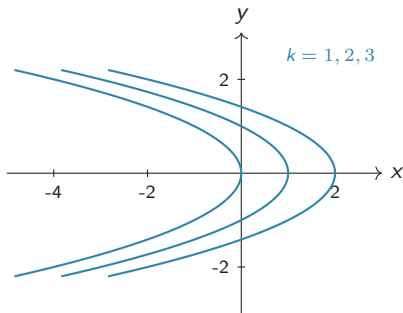
For $f(x, y) = x + y^2 + 1$, the domain is

$$D = \mathbb{R}^2.$$

The level curve of level k is

$$x + y^2 + 1 = k \iff x = k - 1 - y^2.$$

For $k = 1$, this is $x = -y^2$, which passes through the origin.



Question 3

Recall that $\ln(\cdot)$, $\sin(\cdot)$, and $\cos(\cdot)$ are continuous. Prove that the following functions are continuous.

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto (\cos x)^2$.
- (b) $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2, (x, y) \mapsto (\ln(x^4 + 1), \ln(y^4 + 1))$.
- (c) $f: \mathbb{R}^2 \rightarrow \mathbb{R}, (x, y) \mapsto \sin(x^2) + \sin(y^2)$.

Tutorial Q3: Continuity Proofs

Knowledge used

- ▶ Use composition, sums, and componentwise continuity.
- ▶ Projection maps $\text{pr}_i(\mathbf{x}) = x_i$ are continuous.

Solution

(a) Let $g: \mathbb{R} \rightarrow \mathbb{R}$, $t \mapsto t^2$. Then $f = g \circ \cos$, so f is continuous.

(b) Write $f = (f_1, f_2)$, where

$$f_i(\mathbf{x}) = \ln(x_i^4 + 1) = \ln \circ g \circ \text{pr}_i(\mathbf{x}), \quad g(t) = t^4 + 1.$$

Each component is continuous; hence f is continuous.

(c) With $g(t) = t^2$,

$$f = \sin \circ g \circ \text{pr}_1 + \sin \circ g \circ \text{pr}_2.$$

Therefore, f is continuous.

Question 4

The level curves of a production function $Q(K, L)$ are called isoquants. Obtain and plot a few isoquants for the following production functions.

(a) $Q(K, L) = K^{\frac{1}{2}} L^{\frac{1}{2}}$.

(b) $Q(K, L) = 0.5 \ln(K) + 0.5 \ln(L)$.

Tutorial Q4: Isoquants

Knowledge used

- ▶ Isoquants are level curves of a production function.
- ▶ A level curve of level C solves $Q(K, L) = C$.

Solution

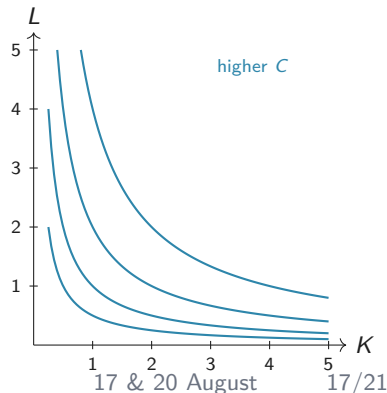
- (a) For $Q(K, L) = K^{1/2}L^{1/2}$, the level curve of level $C > 0$ satisfies

$$K^{1/2}L^{1/2} = C \iff KL = C^2 \iff L = \frac{C^2}{K}.$$

- (b) For $Q(K, L) = 0.5 \ln K + 0.5 \ln L$, the domain is $K, L > 0$. The level curve of level C satisfies

$$0.5 \ln K + 0.5 \ln L = C \iff K^{1/2}L^{1/2} = e^C \iff L = \frac{e^{2C}}{K}.$$

Hence the contour plot has the same shape as part (a).



Question 5

- (a) Write down the formula for a linear function g with domain $\mathbb{R}^2$ and codomain $\mathbb{R}$ that has the property

$$g(7, 2) - g(3, 2) = 8, \quad g(7, 5) - g(3, 2) = 5.$$

- (b) Consider $\hat{g}: \mathbb{R}^2 \rightarrow \mathbb{R}$, $\mathbf{x} \mapsto g(\mathbf{x}) + 2$, where g is the linear function from part (a). How does the value of $\hat{g}$ change if $\mathbf{x}$ changes
- (i) from $(-3, 0)$ to $(2, 1)$?
 - (ii) from $(2, 5)$ to $(1, 3)$?

Tutorial Q5(a): Finding the Linear Function

Knowledge used

Use the lecture definition of a linear function:

$$g(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x} = a_1x_1 + a_2x_2.$$

For linear functions, differences depend only on the change in the input.

Solution

Let $g(\mathbf{x}) = a_1x_1 + a_2x_2$. The two conditions give

$$g(7, 2) - g(3, 2) = 4a_1 = 8 \quad \Rightarrow \quad a_1 = 2,$$

$$g(7, 5) - g(3, 2) = 4a_1 + 3a_2 = 5 \quad \Rightarrow \quad 8 + 3a_2 = 5 \quad \Rightarrow \quad a_2 = -1.$$

Therefore,

$$g(\mathbf{x}) = (2, -1) \cdot \mathbf{x} = 2x_1 - x_2.$$

Tutorial Q5(b): Changes in the Affine Function

Knowledge used

For an affine function $\widehat{g}(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x} + b$, a change $\Delta \mathbf{x}$ gives

$$\Delta y = \mathbf{a} \cdot \Delta \mathbf{x}.$$

The constant term does not affect changes.

Solution

Here $\mathbf{a} = (2, -1)$, so $\Delta y = (2, -1) \cdot \Delta \mathbf{x}$.

(i) From $(-3, 0)$ to $(2, 1)$:

$$\Delta \mathbf{x} = (5, 1),$$

$$\Delta y = (2, -1) \cdot (5, 1) = 9.$$

(ii) From $(2, 5)$ to $(1, 3)$:

$$\Delta \mathbf{x} = (-1, -2),$$

$$\Delta y = (2, -1) \cdot (-1, -2) = 0.$$

Key Takeaways

- ▶ Level curves solve $f(\mathbf{x}) = c$ and contour plots collect several level curves.
- ▶ Continuity can often be proved using sums, products, quotients, compositions, projections, and componentwise arguments.
- ▶ Linear and affine functions convert changes in variables into changes in values through the dot product.